

Cognitive Assessment and Trading Performance Correlations



J. Eduardo Lugo and Jocelyn Faubert

Abstract Neuroeconomics and behavioral finance have provided insight on how cognitive processes and emotions combine to influence financial decisions. In trading decision-making, cognitive assessment and its possible increase through training should be better understood. In this preliminary validation investigation, we employed NeuroTracker (3D multiple object tracking or 3D-MOT), a technique extensively used to test and train cognitive processes in performance populations, to investigate whether the metrics on this task relate to trading performance. The findings demonstrate that there are strong relationships between trading metrics and NeuroTracker scores.

Keywords 3D-MOT · NeuroTracker · Trading · Neuroeconomics · Finance

1 Introduction

There is a case to be made that great sportsmen and gamers share certain mental performance-related challenges with traders. For instance, they must process a lot of information at once, which can lead to cognitive overload, and they may experience performance anxiety, which is known to impair cognitive function [1]. According to the attentional control theory, anxiety decreases the effectiveness of the goal-directed attentional system and increases the extent to which processing is influenced by the stimulus-driven attentional system [2, 3]. When people complete many tasks at once, this impairment may get worse. Numerous studies have shown that performing many activities at once limits a person's ability to understand information [4]. According to Kahneman's [5] theory, this form of "limited attention" enables people to divide their cognitive resources across many activities. As a result, the amount of attention given to one activity must be less than the amount of attention given to other tasks.

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The knowledge that investors may obtain is the focus of earlier research on attention in finance literature. Merton [6], for instance, examined market equilibrium in a situation where investors only had a vague understanding of the securities market. The effects of attention assignment on financial markets and investor attitudes have since been the subject of several theoretical and observational investigations. Peng [7] demonstrates how investors might focus their limited attention on sources of uncertainty to reduce total portfolio uncertainty. According to Peng and Xiong [8], investors with limited attention will utilize fundamental decision-making concepts such as categorization, and these behaviors will clarify well-documented changes in covarying asset returns. Huberman [9], Huberman and Regev [10], Barber and Odean [11] are compatible with these hypotheses, implying that investors prefer known stocks or that information may not be incorporated in pricing until it piques investors' attention. While these studies suggest that there is a lack of attention, accurate assessments are scarce since attention and its relationship to financial markets are difficult to define. Corwin, et al. [4] investigated whether the ability of a specialist to sell shares on the New York Stock Exchange (NYSE) is connected to limited attention. The Hypothesis of Limited Attention is founded on the notion that individual experts suffer time and processing constraints that limit their capacity to follow and manage several requests at the same time, particularly during peak business hours [4]. The most intriguing conclusion was the negative association between a specialist's level of attention to all stocks and the trading operation and the absolute returns of all other stocks in the specialist's portfolio. The data also show that during times of heightened operation, experts focus their efforts on their most productive stocks, resulting in fewer regular market price movements and higher transaction costs for the remaining assigned equities. As a result of limited attention capacity, the attempt distribution has a direct influence on the liquidity provision of financial securities markets.

There is growing evidence that 3D multiple object tracking (3D-MOT) is a powerful and sensitive metric for assessing cognitive-attentional capacity in a variety of populations [12–17]. Tullo et al. [12], for example, studied the resource constraints for dynamic visual attention via growth utilizing a 3D-MOT training interface. According to the findings, perceptual-cognitive functions vary with age, and 3D-MOT may be utilized to characterize resource allocation in attentional processes. Flight, independent of where to practice (simulator or jet), is a cognitively hard task, according to the workload evaluation utilizing perceptual-cognitive approaches. When the 3D-MOT scores at rest were compared to the 3D-MOT scores when the evaluation pilots were flying, the 3D-MOT speed thresholds were dramatically reduced (by an average of 97%). When compared to the simulator arrangement, the results showed that both 3D-MOT and physiological performance were more influenced during a real flight in the aircraft. As a result, it is possible that the cognitive burden in live flight is much higher. These findings confirm the assumption that physiological and environmental noise are greater in real-world settings than in laboratory settings [18], and that brain dynamics fluctuate more in real-world settings than in laboratory settings [19].

The foregoing justifications make it obvious that 3D-MOT is a possible instrument for testing and honing cognitive abilities that are important to traders. Correlating 3D-MOT performance with a secondary job, such as trading performance, would be a first step in this approach. This allows traders to be alerted to any cognitive changes brought on by stress, mental tiredness, or other causes before making poor choices that might cost them money. In the current study, 29 traders' trading metrics and 3D-MOT scores were compared to suggest a novel paradigm for training and testing cognitive function. The findings indicate significant positive or negative associations with trading metrics like total profits or maximum drawdown.

This work is organized as follows: the materials and methods utilized in this work are presented in Sect. 2. We describe our sample selection, the equipment, the technique, the variables of interest, and the statistical approach. Section 3 presents the results and Sect. 4 explores our empirical findings and further analysis, while Sect. 5 provides closing thoughts.

2 Materials and Methods

2.1 *Participants*

NeuroStreet Trading Academy selected 29 traders (all men) aged 35–65 years old (Mean SD = 50 ± 15) for this study. Because this was a retail trader research, only unlicensed independent traders who trade their accounts participated. The data for this study spans from the 26th of Sept 2019–to the 4th Aug 2020. All participants consented to participate in this study and signed a written informed permission to participate. The study followed the principles of the Helsinki Declaration (last modified, 2004).

2.2 *Apparatus*

The NeuroTracker cloud version of the 3D-MOT training program was used in this study (CogniSens Athletics, Inc., Montreal, Quebec, Canada), where the task is to track a specific number of target spheres out of a total of 8 that are observed on a screen, through an expanded 3D virtual visual field [20, 21].

Almost every trader was using the Ninjatrade Trading Platform (Denver, Colorado, USA). Other platforms, such as TradeStation (Plantation, Florida, USA) or Tradingview, were occasionally utilized by traders, however, the platform they traded from had no influence on this study because traders trade from a variety of platforms. The majority of the trading was done in a simulator mode; however, some was done on actual accounts.

Table 1 Relationship between screen size and sit-down distance from the screen

Screen size (inch)	Distance from screen (cm)
65	173
60	160
55	147
50	133
45	120
40	107
35	93
30	80
25	67
20	54
15	40

2.3 Procedure

The study was conducted fully remotely, with participants training using NeuroTracker without supervision. NeuroStreet Trading Academy recommended that all users train either before the trade (typically around 8 a.m. EST) or after the market closes (4 p.m. EST), with just one session done each day. We strongly advised participants to become acquainted with the training program setting and stimuli before to assessment. They were then invited to put on the anaglyph goggles, which allowed them to see the stimuli in 3D. In all training sessions, we instructed traders to sit facing the center of the screen at 44–50” above the ground. To accommodate varying heights, we highly advised the traders to utilize a chair with a seat height adjusting mechanism. The distance from the screen should be changed according to Table 1 depending on the screen size (diagonal).

Traders were instructed to focus their attention on the fixation point, which was presented as a green dot straight ahead in the middle of the screen.

Traders were free to begin trading after completing their 3D-MOT measurement. At the end of their trading trip, the traders electronically reported their trading data each day, indicating whether the data came from a simulator or an actual account and providing their NeuroTracker ratings.

2.4 Measures

Session 3D-MOT speed thresholds were employed to measure changes in cognitive behavior between traders. Total Net Profit, Max Drawdown, were used to analyze trading performance (see Table 2). Trading is an inherently complex endeavor, and

Table 2 The analyzed measurements and the units in which they were recorded are defined. m represents meters, s represents seconds, and USD represents US dollars

Measure	Unit	Description
NeuroTracker Speed threshold	m/s	At the end of each NeuroTracker session, which consists of 20 trials, a speed threshold is determined
Total net profit	USD	The profit made by a corporation or person from their investments before taxes are deducted
Max drawdown	USD	Is the largest observed loss from a portfolio's peak to trough before a new peak is reached. Maximum drawdown is a measure of downside risk over a certain time period

differences in trading competence cannot be fully comprehended solely on a single combination of metrics.

2.5 Statistical Analysis

The association between the 3D-MOT threshold scores and the two trading variables (total net profit, and max drawdown), was initially studied using bivariate correlations using the Pearson technique. All of the variables in our data defied the assumption of normality, according to a preliminary analysis of the data. For all of these variables, non-parametric Spearman correlations were performed instead.

3 Results

3.1 3D-MOT and Trading Scores Correlations

In all, 29 Traders and 624 repeated trading observations are used in this research. These 624 observations will be referred to as Base 1 in the following. Table 3 below presents the summary statistics for this dataset (Base1).

Kolmogorov–Smirnov and Shapiro–Wilk tests for normality were performed. We may reject the null hypothesis that our data is not normal if the significance value

Table 3 The consolidated data for 29 dealers

Score	N	Minimum	1st quartile	Median	3rd quartile	Maximum	Mean	Std. deviation
3D-MOT	624	0.01	0.76	1.19	2.69	5.13	1.70	1.44
TotalNetProfit	624	−630.00	− 0.52	156.25	842.37	131,367.64	3500.10	12,683.41
MaxDrawdown	624	−2,329.72	− 775	− 150	0.00	0.00	−1010.28	2241.41

Table 4 Spearman’s correlations results

		Score
Score		3D-MOT
3D-MOT	Correlation coefficient	1.000
	Sig. (2-tailed)	–
	N	624
TotalNetProfit	Correlation coefficient	0.487**
	Sig. (2-tailed)	0.000
	N	624
MaxDrawdown	Correlation coefficient	–0.323**
	Sig. (2-tailed)	0.000
	N	624

The significance of asterisk in the **p<0.001

exceeds the alpha value of 0.05, which means that there is no reason to believe that our data deviates considerably from a normal distribution. We found that both tests produce significant values that are less than .05, indicating that we may be sure that our data are not normally distributed. For these variables, non-parametric correlations were thus performed. Using Base 1, Spearman correlations were carried out. 3D-MOT scores were shown to be significantly correlated with Total Net Profit, and Max Drawdown. Max Drawdown had a negative correlation with 3D-MOT scores, and for TotalNetProfit it was positive. These findings are shown in Table 4.

4 Discussion

One crucial distinction to make is that, in prior 3D-MOT investigations, the task was carried out in a lab or under supervision, but the current research was conducted in the wild with no monitoring of the individuals. Therefore, it is presumed that they correctly followed the directions given.

The current findings demonstrate a distinct relationship between trading success and the equivalent basic mental abilities for mastering an abstract and challenging dynamic scene job like the 3D-MOT. There was a positive statistically significant relationship between Total Net Profit. This trading metric corresponds to earnings, commissions, or profits. Meaning that when 3D-MOT scores rise, the metrics Total Net Profit, tend to rise as well. With Total Net Profits and 3D-MOT scores, this association is more pronounced (near to 50%). Max Drawdown losses tend to rise when 3D-MOT scores fall, hence the correlation is reversed for Max Drawdown. The second-highest correlation we identified is this one (almost –32%). This finding is significant since a decline in cognitive function as determined by 3D-MOT coincides with a loss trading parameter. It doesn’t matter where the drop in cognitive

performance comes from—anxiety, mental exhaustion, or in severe situations, physical concussions—what matters is that it can be quantified and connected with a secondary task, in this case, a negative trading characteristic. The inverse is also true; an improvement in cognitive performance as measured by 3D-MOT substantially corresponds with favorable trading characteristics.

5 Conclusions

According to the current feasibility research, cognitive capabilities as assessed by the 3D-MOT paradigm, which have been linked to performance levels in top sports and professional video gamers, also seem to correspond with performance indicators in traders. Significant connections between 3D-MOT scores and performance-related trading factors were discovered. We may draw the conclusion that the 3D-MOT measure has the potential as a useful tool for controlling and developing performance-related cognitive processes crucial for the trading profession based on the association between the 3D-MOT measures and the performance metrics of traders displayed herein. In the future, these correlations will help managers, traders, investors, and brokers avoid lower cognitive performance that might be brought on by a number of reasons, including fatigue or performance anxiety, and hence help them make better investment decisions.

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